



MID SEMESTER EXAMINATION, AUTUMN-2024
School of Applied Sciences, KIIT-DU

B. Tech.
1st Semester (2024AB)
Autumn 2024 (SAS)

Course: Environmental Sciences	Code: CH10003	Full Marks: 20	Time: 90 minutes
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Answer any FOUR QUESTIONS including question No. 1 which is compulsory.

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

All parts of a question should be answered at one place only.

Q.1		Marks	CO
a)	Why atmosphere is important to the habitats?	1	1
b)	What is the difference between convection and conduction during thermal distribution?	1	1
c)	State the Stefan-Boltzmann theory of radiation.	1	1
d)	What are the positive impacts of carbon dioxide to the nature?	1	2
e)	What is the main purpose of the Montreal Act?	1	1

Q.2		Marks	CO
a)	Describe in short about the stratosphere and thermosphere? What are the positive and negative impacts of stratosphere and thermosphere in to that of the atmosphere and to that of the Earth surface?	2.5	1
b)	What are organic particulate matters? How PAH is formed from alpha-benzopyrene-sketch in details with adverse to the environment/public health?	2.5	1

Q.3		Marks	CO
a)	CFC in conjugation with UV-light responsible to destroy the ozone layer-mechanistically proof this? Write down the structural formula of CFC-12.	3	1
b)	Why the Global warming is caused? What are the Pros and Cons of the Global warming to the environment?	2	1

Q.4		Marks	CO
a)	Mechanistically explain the formation of oxidizing smog with its adverse effect to the environment.	3	2
b)	Why acid rain is so toxic? What are the adverse effects of acid rain to the environment?	2	2

Q.5		Marks	CO
a)	An imaginary black body is heated at 6000K. If the Boltzmann constant value is $1.3806452 \times 10^{-23}$ J/K, then calculate the energy gradient change with respect to the wavelength.	3	1
b)	What does it mean by lithosphere? Do segment the lithosphere? What are the Pros and Cons of top soil for the plants regarding water collection for photosynthesis?	2	2
